t3-w02 — Plane Shapes (2-D Shapes) — Plane shapes: types, properties and perimeter

Run `diag-v3` · conditions C0 and C1 · the pack shown is draw 3 of each.

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Plane Shapes (2-D Shapes)

`subtopic`

Plane shapes: types, properties and perimeter

`term`

Third Term

`week`

2

`lesson_objectives`

• identify the types of plane shapes and state their properties, similarities and differences
• identify the parts of a circle
• find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Objective 1 — identify the types of plane shapes and state their properties, similarities and differences

`objective_text`

identify the types of plane shapes and state their properties, similarities and differences

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — identify the parts of a circle

`objective_text`

identify the parts of a circle

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 3 — find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`objective_text`

find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

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Output — the pack it produced (draw 3)

Objective 1

Types of Plane Shapes and Their Properties

> EXPLANATION
> A **plane shape** (or 2-D shape) lies flat on a surface and has length and width but no depth. The most common families are:
>
> - **Triangle** – 3 sides, 3 angles. Types include equilateral (all sides equal, all angles 60°), isosceles (two sides equal), and scalene (no sides equal).
> - **Quadrilateral** – 4 sides, 4 angles. Important kinds are:
> - **Square** – 4 equal sides, 4 right angles.
> - **Rectangle** – opposite sides equal, 4 right angles.
> - **Parallelogram** – opposite sides equal and parallel, opposite angles equal.
> - **Rhombus** – 4 equal sides, opposite angles equal (angles need not be right).
> - **Trapezium (trapezoid)** – at least one pair of parallel sides.
> - **Pentagon** – 5 sides, 5 angles. A *regular* pentagon has all sides and angles equal.
> - **Hexagon** – 6 sides, 6 angles. A *regular* hexagon has all sides and angles equal.
> - **Circle** – no straight sides, defined by a centre and a radius; its perimeter is called the circumference.
>
> **Similarities**: All polygons (triangle, quadrilateral, pentagon, hexagon, etc.) are made of straight line segments that join end-to-end, and the sum of their interior angles follows the rule \((n-2)\times180° \) where \(n\) is the number of sides. Squares, rectangles, and rhombuses are all *quadrilaterals* and share the property of having opposite sides parallel.
>
> **Differences**: The number of sides and angles distinguishes each family. Regular polygons have all sides and angles equal, whereas irregular ones do not. Circles differ from all polygons because they have no straight sides and are described by radius rather than side length.
>
> Understanding these properties helps you recognise a shape quickly and compare it with others.

> WORKED_EXAMPLE
> **Worked Example**
>
> *Problem*: A shape has four sides of equal length and each interior angle measures 90°. Identify the shape and list three of its properties.
>
> *Solution*: Because the shape has four equal sides, it could be a square or a rhombus. The fact that each interior angle is 90° tells us the shape is a **square** (a rhombus does not require right angles). Three properties of a square are:
> 1. All four sides are equal.
> 2. All four interior angles are right angles (90°).
> 3. Opposite sides are parallel.
>
> *Comparison*: A rectangle also has four right angles, but its adjacent sides can be of different lengths. Thus, the square and rectangle are similar in having four right angles and opposite sides parallel, but they differ in side-length equality.

1. 1. Identify the shape that has exactly three sides and name one of its special types.
answer: Triangle; a special type is an equilateral triangle.

2. 2. Which shape has four equal sides and four right angles? State two properties it shares with a rectangle.
answer: Square; it shares that opposite sides are parallel and that all interior angles are 90°.

3. 3. List two ways a regular pentagon and a regular hexagon are similar.
answer: Both are regular polygons (all sides equal and all interior angles equal) and both have all vertices lying on a single circle (they are cyclic).

4. 4. A shape has one pair of parallel sides and the other two sides are not parallel. What is the shape called, and what is a key property of its angles?
answer: Trapezium (trapezoid); the two interior angles adjacent to each parallel side are supplementary (add to 180°).

5. 5. True or False: All circles are regular polygons.
answer: False – a circle is not a polygon because it has no straight sides.

Objective 2

Parts of a Circle

EXPLANATION

A **circle** is a set of points that are all the same distance from a fixed point called the **centre**. The key parts of a circle are:

- **Radius** – a line segment from the centre to any point on the circle.
- **Diameter** – a line segment that passes through the centre and whose endpoints lie on the circle; it is twice the length of the radius.
- **Chord** – any line segment whose endpoints are on the circle. If a chord passes through the centre, it is a diameter.
- **Arc** – a part of the circumference (the outer edge) between two points on the circle.
- **Sector** – the region bounded by two radii and the arc between them; it looks like a ‘slice of pizza’.
- **Central angle** – the angle formed at the centre by two radii.
- **Circumference** – the complete distance around the circle.

Understanding these parts helps you describe and work with circles in geometry.

WORKED_EXAMPLE

Worked Example

Imagine a circle with centre **O**. A line segment **OA** is drawn from **O** to a point **A** on the circle.

- What is **OA**? It joins the centre to the edge, so it is a **radius**.

Now draw another line segment **OB** that also joins the centre to a different point **B** on the circle.

- The segment **AB** joins two points on the circle but does **not** pass through the centre. This is a **chord**.

If we draw a line through the centre that meets the circle at points **C** and **D**, the segment **CD** is the longest possible chord, so it is the **diameter**.

The part of the circle’s edge from **A** to **B** is called an **arc** (specifically, the minor arc AB).

Finally, the region enclosed by the two radii **OA**, **OB**, and the arc **AB** is a **sector** of the circle.

This example shows how each term describes a specific part of the circle.

1. 1. The line segment from the centre of a circle to any point on the circle is called the _____.
answer: radius

2. 2. The longest chord of a circle, passing through the centre, is the _____.
answer: diameter

3. 3. A line segment that joins two points on the circle but does not pass through the centre is a _____.
answer: chord

4. 4. The part of the circle’s edge between two points on the circle is called an _____.
answer: arc

5. 5. The region bounded by two radii and the included arc is known as a _____.
answer: sector

Finding Perimeters of Common Shapes

EXPLANATION

The perimeter of a shape is the total distance around its edge.

Formulas

- Regular polygon with n equal sides of length s: $P = n \times s$.
- Square with side a: $P = 4a$.
- Rectangle with length l and width w: $P = 2(l + w)$.
- Trapezium (trapezoid) with side lengths a, b, c, d (the two parallel sides are a and b): $P = a + b + c + d$.
- Parallelogram with base b and side s: $P = 2(b + s)$.
- Circle (circumference) with radius r (or diameter $d = 2r$): $C = 2\pi r = \pi d$. Use $\pi \approx 3.14$ for calculations.

WORKED_EXAMPLE

Worked examples

1. *Regular pentagon*: each side = 6 cm. $P = 5 \times 6 = 30$ cm.
2. *Square*: side = 5 cm. $P = 4 \times 5 = 20$ cm.
3. *Rectangle*: length = 8 cm, width = 3 cm. $P = 2(8 + 3) = 22$ cm.
4. *Trapezium*: sides = 4 cm, 5 cm, 6 cm, 7 cm. $P = 4 + 5 + 6 + 7 = 22$ cm.
5. *Parallelogram*: base = 9 cm, side = 4 cm. $P = 2(9 + 4) = 26$ cm.
6. *Circle*: radius = 3 cm. $C \approx 2 \times 3.14 \times 3 = 18.84$ cm.

1. A regular hexagon has each side 4 cm long. What is its perimeter?
answer: 24 cm

2. A square has a perimeter of 36 cm. What is the length of each side?
answer: 9 cm

3. A rectangle has length 12 cm and width 5 cm. Find its perimeter.
answer: 34 cm

4. A trapezium has parallel sides 8 cm and 5 cm, and the other two sides are 4 cm and 6 cm. Find its perimeter.
answer: 23 cm

5. A parallelogram has a base of 7 cm and a side length of 3 cm. What is its perimeter?
answer: 20 cm

6. A circle has a diameter of 10 cm. Find its circumference (use $\pi \approx 3.14$).
answer: 31.4 cm

C1 The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

subject

Mathematics

grade_level

JSS1

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Plane Shapes (2-D Shapes)

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Plane shapes: types, properties and perimeter

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lesson_objectives

• identify the types of plane shapes and state their properties, similarities and differences
• identify the parts of a circle
• find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

entry_behaviour

• Students have been taught angles.

instructional_materials

• Cardboard containing angles presentation.

Objective 1 — identify the types of plane shapes and state their properties, similarities and differences

`objective_text`

identify the types of plane shapes and state their properties, similarities and differences

`current_step`

1

`instructional_events`

- present_content
- elicit_performance
- present_content

`assessment_points`

- How many triangles has an octagon.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — identify the parts of a circle

`objective_text`

identify the parts of a circle

`current_step`

2

`instructional_events`

- present_content

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 3 — find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`objective_text`

find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

`current_step`

3

`assessment_points`

• Find the perimeter of a square of sides 8cm
• Find the perimeter of a rectangle of length 7cm and breadth 4cm.
• What is the perimeter of regular pentagon with sides 5cm.
• Find the perimeter of a parallelogram whose base is 6cm and sides 5cm
• The circumference of a circle is 22cm – Find the radius of a circle $(\pi = \frac{22}{7})$.
• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle.
• Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm]
• Find the perimeter of a regular hexagon with sides 5cm.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `instructional_events`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Identifying Plane Shapes and Their Properties

EXPLANATION

A **plane shape** (or 2-D shape) is a flat figure that lies in a single plane. All plane shapes are closed, have a perimeter, and enclose an area.

Common types of plane shapes

- **Triangle** – 3 sides, 3 vertices, interior angles sum to 180°.
- **Quadrilaterals** – 4 sides, 4 vertices, interior angles sum to 360°. Important kinds:
  - *Square*: all sides equal, all angles 90°.
  - *Rectangle*: opposite sides equal, all angles 90°.
  - *Rhombus*: all sides equal, opposite angles equal (not necessarily 90°).
  - *Parallelogram*: opposite sides parallel and equal, opposite angles equal.
  - *Trapezium* (trapezoid): at least one pair of parallel sides.
- **Pentagon** – 5 sides, interior angles sum to 540°.
- **Hexagon** – 6 sides, interior angles sum to 720°.
- **Octagon** – 8 sides, interior angles sum to 1080°.
- **Circle** – no straight sides, defined by a centre and radius; perimeter is called the circumference.

Similarities
- All are closed figures.
- All have a perimeter that can be measured.
- All enclose an area.
- All can be regular (all sides and angles equal) or irregular.

Differences
- Number of sides and vertices.
- Size of interior angles.
- Whether opposite sides are parallel.
- Whether all sides are equal.
- Presence or absence of curved edges (circle).

WORKED_EXAMPLE

Worked Example – Comparing a Square and a Rectangle

Both a square and a rectangle are quadrilaterals, so they share these properties:
- Four sides and four vertices.
- Opposite sides are parallel.
- The sum of the interior angles is 360°.
- All interior angles are right angles (90°).

What makes a square different?
- In a square, *all four sides are equal* in length.
- In a rectangle, only *opposite sides* are equal; the adjacent sides can be of different lengths.

Example: If a square has side length 5 cm, its perimeter is $4 \times 5 = 20$ cm. A rectangle with length 8 cm and width 5 cm also has right angles, but its perimeter is $2 \times (8 + 5) = 26$ cm because the side lengths are not all the same.

1. 1. List three different types of quadrilaterals and state one property that each of them shares.
answer: Square, rectangle, rhombus – all have opposite sides parallel and equal.

2. 2. How many sides does a regular pentagon have, and what is the measure of each interior angle?
answer: 5 sides; each interior angle is 108°.

3. 3. Which of the following shapes has at least one pair of parallel sides? (a) triangle (b) square (c) circle (d) pentagon
answer: (b) square

4. 4. True or False: All rectangles are squares.
answer: False

5. 5. What is a key difference between a regular hexagon and a regular octagon?

answer: A regular hexagon has 6 sides with each interior angle 120°, while a regular octagon has 8 sides with each interior angle 135°.

Objective 2

Parts of a Circle

EXPLANATION

A circle is a set of points that are all the same distance from a fixed point called the centre. The main parts of a circle are:

- **Centre** – the fixed point in the middle.
- **Radius** – a line segment from the centre to any point on the circle.
- **Diameter** – a line segment that passes through the centre and whose endpoints lie on the circle; it is twice the radius.
- **Circumference** – the distance around the circle; its length is $C = 2\pi r$ or $C = \pi d$.
- **Chord** – a line segment whose endpoints are on the circle but does not necessarily pass through the centre.
- **Arc** – a part of the circumference between two points on the circle.
- **Sector** – the region bounded by two radii and the arc between them.
- **Segment** – the region bounded by a chord and the arc it subtends.

These parts are useful when describing positions on a circle or calculating distances.

WORKED_EXAMPLE

Worked Example: Find the circumference of a circle whose diameter is 12 cm.

1. First find the radius: $r = \frac{d}{2} = \frac{12}{2} = 6$ cm.
2. Use the formula $C = 2\pi r$.
   $C = 2 \times 3.14 \times 6 = 37.68$ cm.

So the circumference is 37.68 cm.

1. What is the name of the line segment that joins the centre of a circle to any point on the circle?
answer: radius

2. A circle has a diameter of 14 cm. What is its radius?
answer: 7 cm

3. Calculate the circumference of a circle with radius 5 cm. Use $\pi \approx 3.14$.
answer: 31.4 cm

4. What do we call the part of the circle's edge between two points on the circle?
answer: arc

Perimeter Practice Pack

> EXPLANATION
> The perimeter of a shape is the total length around it. For regular polygons each side has the same length, so perimeter = number of sides × side length. For a square, 4 equal sides → P = 4a. For a rectangle, two lengths and two widths → P = 2(l + w). For a parallelogram, two equal pairs → P = 2(base + side). For a trapezium, add the lengths of all four sides. For a circle, the distance around is the circumference C = 2πr = πd. With π = 22/7 you can find the radius from the given circumference.

> WORKED_EXAMPLE
> Example: Find the perimeter of a square with side 8 cm. Since a square has four equal sides, P = 4 × 8 cm = 32 cm. Example: Find the perimeter of a rectangle 7 cm by 4 cm. P = 2(7 cm + 4 cm) = 2 × 11 cm = 22 cm.

> WORKED_EXAMPLE
> Example: A regular pentagon has five sides each 5 cm. P = 5 × 5 cm = 25 cm. Example: The circumference of a circle is 22 cm. Using π = 22/7, C = 2πr ⇒ r = C / (2π) = 22 cm / (2 × 22/7) = 22 cm / (44/7) = 22 cm × 7/44 = 3.5 cm.

1. Find the perimeter of a square of sides 8 cm.
answer: 32 cm

2. Find the perimeter of a rectangle of length 7 cm and breadth 4 cm.
answer: 22 cm

3. What is the perimeter of a regular pentagon with sides 5 cm?
answer: 25 cm

4. Find the perimeter of a parallelogram whose base is 6 cm and sides 5 cm.
answer: 22 cm

5. The circumference of a circle is 22 cm – Find the radius of the circle (π = 22/7).
answer: 3.5 cm

6. A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle.
answer: 6d

7. Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm]
answer: 15 cm

8. Find the perimeter of a regular hexagon with sides 5 cm.
answer: 30 cm

Result — per rubric dimension

Scores are 1–5, each the mean of 2 judge pass(es) on the same packet. The first three columns are the packs shown above (draw 3); the last is the lesson's mean over its 3 draws per condition — the study's number.

Rubric	C0	C1	C1 – C0, shown packs	C1 – C0, mean of 3 draws
Objective and curriculum alignment	5.00	5.00	+0.00	+0.00
Taught-lesson alignment	4.00	4.00	+0.00	**+0.17**
Scope and conceptual correctness	4.00	4.00	+0.00	**+0.17**
Student-facing clarity and usefulness	5.00	4.50	**-0.50**	**-0.33**